

Deep Generative Models

Chapter 6: Generation by Diffusion Process

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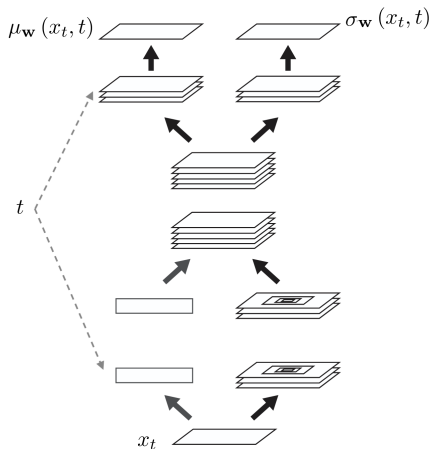
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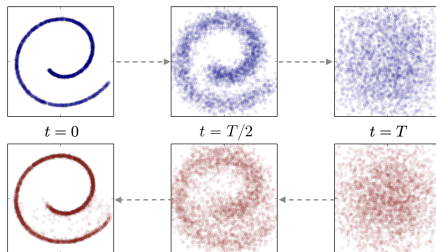
DPM: Sohl-Dickstein et al.

Sohl-Dickstein et al. implemented the first DPM in their work

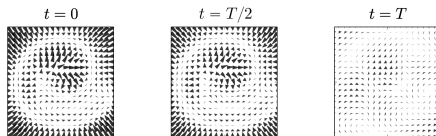


DPM: Sohl-Dickstein et al.

They tried it on the Swiss roll dataset to demonstrate the diffusion process

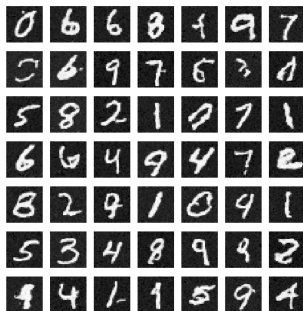


Interestingly, they could see the drift behavior



DPM: Sohl-Dickstein et al.

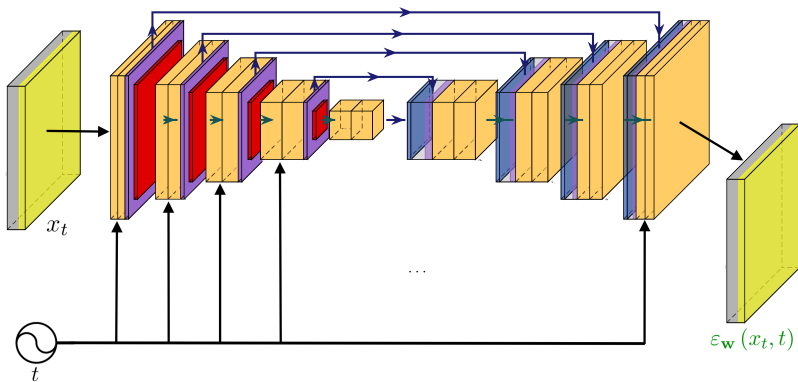
They also tried couple of classical datasets including MNIST and CIFAR



But, it was not the beating the benchmark at the time, i.e., 2015

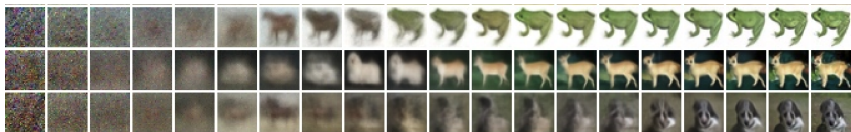
DDPM: Ho et al.

Ho et al. proposed DDPM and implemented it by embedding time in U-Net



DDPM: Ho et al.

They could demonstrate efficient reverse diffusion



and very inspiring high-quality samples



Improved DDPM: Nichol and Dhariwal

Nichol and Dhariwal kept U-Net, but improved training procedure by

- learning the **variances** for the **reverse** diffusion, and
- tuning the **time scheduling** for the noising process, i.e., α_t

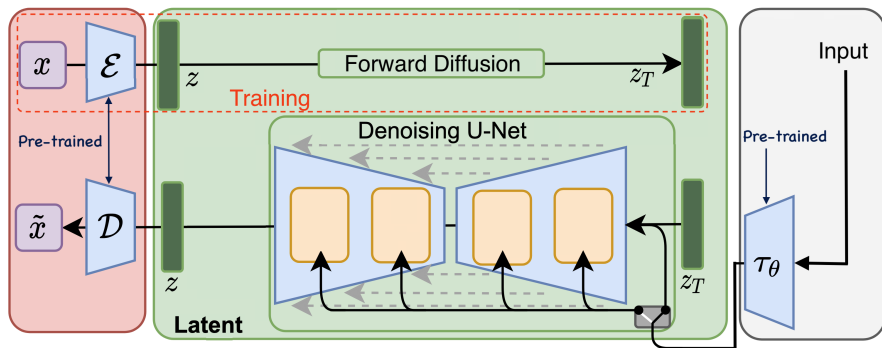
which ended up better sampling quality

Model	ImageNet	CIFAR
Glow (Kingma & Dhariwal, 2018)	3.81	3.35
Flow++ (Ho et al., 2019)	3.69	3.08
PixelCNN (van den Oord et al., 2016c)	3.57	3.14
SPN (Menick & Kalchbrenner, 2018)	3.52	-
NVAE (Vahdat & Kautz, 2020)	-	2.91
Very Deep VAE (Child, 2020)	3.52	2.87
PixelSNAIL (Chen et al., 2018)	3.52	2.85
Image Transformer (Parmar et al., 2018)	3.48	2.90
Sparse Transformer (Child et al., 2019)	3.44	2.80
Routing Transformer (Roy et al., 2020)	3.43	-
DDPM (Ho et al., 2020)	3.77	3.70
DDPM (cont flow) (Song et al., 2020b)	-	2.99
Improved DDPM (ours)	3.53	2.94

Stable Diffusion: *First Large-Scale LDM*

Stable Diffusion employs a VAE to get into the latent space

↳ It uses diffusion to generate the latent representation of data



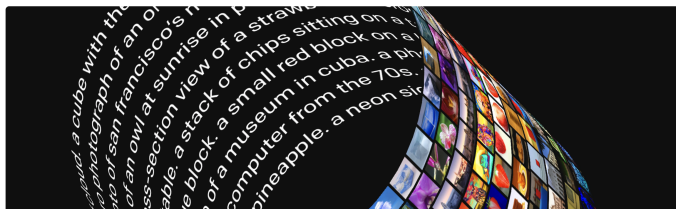
This is a *latent diffusion model* (LDM)

DALL-E: Another LDM

OpenAI also developed an LDM which uses diffusion for image generation

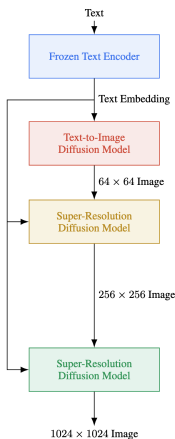
January 5, 2021 Milestone

DALL-E: Creating images from text

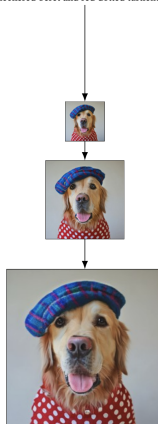


Google Imagen: *Diffusion in Image Space*

Google developed *Imagen* which uses diffusion in the image space



"A Golden Retriever dog wearing a blue checkered beret and red dotted turtleneck."



Final Notes

Probabilistic diffusion learns the *reverse* diffusion process

- *DPMs* learn the reverse diffusion by *ELBO maximization*
- *DDPMs* are special case of DPMs which learn to *estimate noising process*
- Sampling diffusion models can take *too long*
 - ↳ We need to move *enough number of time samples*, e.g., $T = 1000$
 - ↳ We can reduce sampling time by *tricks like DDIM*, e.g., down to $T = 50$

Diffusion models scale *slower* than other models with data complexity

- + We can learn to sample *complex* data *more efficiently*
 - ↳ We only need to learn how to *denoise* through time
 - ↳ Data complexity does not involve *directly* in the process
- They are *overkill* for simple data
 - ↳ We can learn to sample MNIST *much easier* via GANs or VAEs
 - ↳ For simple data, we better *stick to easier models*